Unit 1

Atomic Structure

Topic Breakdown

1.1	–	Atomic Structure	Due Date: _____
1.2	–	Atomic History	Due Date: _____
1.3	–	Isotopes and Atomic Mass	Due Date: _____
1.4	–	Avogadro's Number	Due Date: _____

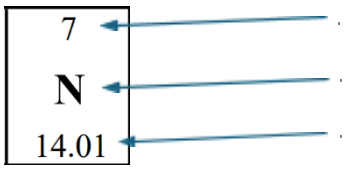
Test Date: Wednesday, September 3rd

To be eligible for retakes, all homework must be completed and submitted. If you have any missing assignments for that unit, you will not be permitted to retake the assessment.

Topic homework must be scanned as a PDF and uploaded to Schoology by the assigned due date. Submissions will be locked once the deadline has passed. If you miss the deadline, you may earn up to 50% credit by showing your work to me in person. This will allow you to maintain your eligibility for a retake.

1.1 Atomic Structure and Properties

1. What are the 3 subatomic particles present in an atom? _____
2. Briefly describe the location of each subatomic particle that you listed above. Be specific about where they are found.
   a. _____
   b. _____
   c. _____
3. Which particle controls the identity of the atom? _____
4. Which particle(s) control the mass of the atom? _____
5. Which particle(s) control the overall charge of the atom? _____
6. Label the parts of the following diagram



7. Fill in the table below with the appropriate information. **Assume all atoms contain a neutral charge.**

Isotope Name	**Nuclear Symbol**	**Atomic Number**	**Mass Number**	**# of Protons**	**# of Electrons**	**# of Neutrons**
1. calcium-40						
2.		12	24			
3.				1		2
4.	$^{197}_{79}Au$					
5.					26	30
6.			201	80		
7.		17				18

8. How does an atom form an ion? _____
9. What are the two types of ions? _____

10. Identify each of the following as one of the types of ions you listed above. List out the number of protons and electrons for each.
    a. Mg^{+2} _____ p^+ : ____ e^- : ____
    b. F^- _____ p^+ : ____ e^- : ____
    c. S^{-2} _____ p^+ : ____ e^- : ____
    d. Li^+ _____ p^+ : ____ e^- : ____

1.2 – Atomic History

1. When 24.0 g magnesium burns in the presence of oxygen according to the reaction below, the final product has a mass of 40.0 grams. Calculate the amount of oxygen gas that was reacted.

$$2Mg + O_2 \rightarrow 2MgO$$

24 g ____g 40.0 g

   a. Calculate the amount of oxygen that reacted with the magnesium.

   b. Explain your answer above using one Dalton's Atomic Theory.

2. Dalton believed that atoms were the smallest form of matter and that they were indivisible. Refute that idea with modern atomic theory.

3. J.J. Thomson was an English scientist who performed research using a cathode ray tube (Crooke's tube).
   a. What major subatomic particle did J.J. Thomson discover? _____
   b. J.J. Thomson determined that this subatomic particle had a charge. What evidence did J.J. Thomson use to determine the charge of the particle he discovered?

   c. J.J. Thomson developed a model for the atomic based on his discovery. Write the name for the model and draw it below. Provide a brief description of the model.



Model Name: _____

Brief Description:

4. Ernest Rutherford conducted a famous experiment using gold foil and a beam of alpha particles produced by a radioactive source of radium. Alpha particles are positively charged Helium atoms that are ejected from a radioactive element. Thomson believed that these alpha particles had enough energy to pass through atoms.
   a. The alpha particles were predicted to pass right through atoms based on the model developed by J.J. Thomson. Why did Rutherford believe this? Remember that positive charges repel each other. Why would Rutherford expect a positive alpha particle to travel right through the positive charge of the plum pudding model?

   b. Discuss the results of the experiment. Be specific about what occurred to the alpha particles.

   c. Based on the experiment, describe what Rutherford proposed about the atom and draw his model that he developed.



Topic 1.3 – Isotopes and Atomic Mass

1. What is an isotope?

2. Why are the atomic masses of each element typically a decimal?

3. Carbon has 3 known naturally occurring isotopes: C-12, C-13, and C-14. The table provided below indicates the abundance for each known isotope.

C-12	98.9%
C-13	1.1%
C-14	Trace

   a. A student obtains 1000 atoms of a carbon sample. Assume that the sample contains no atoms of C-14. How many atoms of C-12 and C-13 are assumed to be present in the sample of carbon?
   C-12: _____ C-13: _____

   b. Explain why the atomic mass of carbon is closer to C-12 than C-13 or C-14.

4. A student is analyzing a sample of an unknown element X. Element X has 3 isotopes with the abundances provided below in the data table. Calculate the atomic mass of element X and determine the identity of the element.

X-29	35.5%
X-30	42.6%
X-31	21.9%

5. Calculate the average atomic mass for the following element given information about the relevant isotopes. Identify unknown element.
   (Isotope 1) Mass = 20 amu; Isotopic abundance = 90.92%
   (Isotope 2) Mass = 21 amu; Isotopic abundance = 0.257%
   (Isotope 3) Mass = 22 amu; Isotopic abundance = 8.82%

6. In the world of chemistry, atoms are so small that we need a unit to quantify a collection of atoms, just like we use a dozen to quantify items.
   a. What is the unit we use to quantify atoms? _____
   b. What is the relationship between the number of atoms and the unit you answered above?

7. A student obtains 1 mole of element X (from problem number 4).
   a. How many atoms of X are present?

   b. The student wishes to calculate the number of atoms of each isotope present in 1 mole of element X. Use the percentages in the data table in problem 4 to help.

      Atoms of X-29 : _____

      Atoms of X-30: _____

      Atoms of X-31: _____

Topic 1.4 – Avogadro's Number and Molar Mass

1. In the world of chemistry, atoms are so small that we need a unit to quantify a collection of atoms, just like we use a dozen to quantify items.
    a. What is the unit we use to quantify atoms? _____
    b. What is the relationship between the number of atoms and the unit you answered above?

2. A student obtains 1 mole of element X (from problem number 4).
    a. How many atoms of X are present?

    b. The student wishes to calculate the number of atoms of each isotope present in 1 mole of element X. Use the percentages in the data table in problem 4 to help.

    Atoms of X-29 : _____

    Atoms of X-30: _____

    Atoms of X-31: _____

3. For the following elements, how many grams are present in 1 mole of each substance?
    a. Carbon: _____
    b. Oxygen: _____
    c. Neon: _____
    d. Sodium: _____
    e. Potassium: _____
    f. Phosphorus: _____

4. Using avogadro's number, perform the following conversions.

    a. Convert 2.50 moles of carbon to atoms of carbon. An outline for the dimensional analysis has been started for you.

$$2.50 \text{ mol C} \times \frac{_____ \text{ atoms c}}{_____ \text{ mol c}} =$$

    b. Convert 1.50 moles of lithium to atoms of lithium.

    c. Convert 5.15×10^{23} atoms of sulfur to moles of sulfur

    d. Convert 1.25×10^{24} atoms of silicon to moles of silicon.